

CHUKA UNIVERSITY
DEPARTMENT OF COMPUTER SCIENCE
COSC 0140: FUNDAMENTALS OF COMPUTER PROGRAMMING (L/P 30/30; CF 3.0)
COURSE OUTLINE

Course Purpose

To introduce students in computer programming

Expected Learning Outcomes

At the end of the course, students should be able to;

- i. Appreciate the role of computer programming languages in problem solving.
- ii. Discuss various steps a computer solution to a problem undergoes.
- iii. Appreciate computer programming concepts using a high level programming language.
- iv. Interpret and design solutions to problems that require decision making and iteration using various problem solving techniques.
- v. Write Interpret, compile and run programs.

Course Content

Overview of programming languages, evolution, assemblers, linkers, and loaders, compilers, interpreters. Algorithms, flowcharts, pseudocode; Introduction to programming methodology and problem-solving strategies: the role of algorithms in the problem-solving process; implementation strategies for algorithms; debugging strategies; the concept and properties of algorithms. Simple data types. Constants and variable declaration. Arithmetic and logical operators. Use of arrays. Program development using a higher level programming language such as PASCAL or Python. Use of control structures. Functions and procedures. Working with text files. Programming techniques: top down methodology, design, code, debug, & program documentation.

Week	Topic	Sub Topic
Week 1	Overview of programming languages	Evolution, assemblers, linkers, and loaders, compilers, interpreters
Week 2	Algorithms, flowcharts, pseudocode	Writing algorithms, pseudocode, drawing flowcharts of different programs
Week 3	Introduction to programming methodology and problem-solving strategies	The role of algorithms in the problem-solving process, implementation strategies for algorithms
Week 4	Debugging strategies	Program errors, how to debug the program
Week 5	CAT 1	
Week 6	Python language introduction	Simple data types. Constants and variable declaration. Arithmetic and logical operators
Week 7	Arrays	Use of arrays, Arithmetic and logical operators, Control structures.
Week 8	Programming techniques	Programming techniques: top down methodology, design, code, debug
Week 9	CAT 2	
Week 10	Wrap up	CAT and assignment feedback Course wrap up, program documentation

Teaching Methods:

Lectures, 2 Hours per week Practical: 2 Hours per week

Course Evaluation: Final Examination 70%; Continuous Assessment 30% , Total: 100%

References

- i. Savitch, W. J. (2000). Java: an introduction to computer science and programming.
- ii. Prentice Hall PTR. Siegel, J. (2000). Computer fundamentals and programming (Vol. 2). Chichester:
- iii. John Wiley & Sons. Horstmann, C. S., & Cornell, G. (2002). Core Java 2: Volume I, Fundamentals. Pearson Education. Pattis, R. E. (1981).
- iv. Karel the robot: a gentle introduction to the art of programming.
- v. John Wiley & Sons, Inc.. Leeds, H. D., & Weinberg, G. M. (1970). Computer programming fundamentals.

Lecturer: Mercy Njeru. Contacts mercynjeru.w@gmail.com